



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 10

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FIRST REVISION EXAMINATION - JANUARY - 2025

Time Allowed : 3.00 Hours]

SCIENCE

[Max. Marks : 75

PART - I

I. Choose the correct answer.

12x1=12

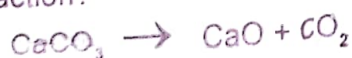
- If the Earth shrinks to 50% of its real radius its mass remaining the same, the weight of a body on the earth will
 - Decrease by 50%
 - Increase by 50%
 - Decrease by 25%
 - Increase by 300%
- In Myopic eye, the image of the object is formed
 - Behind the Retina
 - On the Retina
 - In front of the Retina
 - On the blind Spot
- The Value of Boltzmann constant is
 - $1.381 \times 10^{-23} \text{ JK}^{-1}$
 - $1.381 \times 10^{23} \text{ JK}^{-1}$
 - $1.8 \times 10^{-23} \text{ JK}^{-1}$
 - $8.31 \times \text{J mol}^{-1} \text{ K}^{-1}$
- Kilowatt is the unit for
 - Resistivity
 - Conductivity
 - Electrical Energy
 - Electrical power
- Which of the Following is a Poly atomic molecule?
 - Glucose
 - helium
 - Carbondioxide
 - Hydrogen
- is an important metal to form amalgam
 - Ag
 - Hg
 - Mg
 - Al
- Which of the following is the Universal Solvent?
 - Acetone
 - Benzene
 - Water
 - Alcohol
- Which is formed during Anaerobic Respiration.
 - Carbohydrate
 - Ethyl Alcohol
 - Acetyl COA
 - Pyruvate
- Which is called as Heart of Heart?
 - SA node
 - AV node
 - Purkinje fibre
 - Bundle of His
- The Hormone which has positive effect on apical dominance is
 - Cytokinin
 - Auxin
 - Gibberellin
 - Ethylene
- What is true of Gametes?
 - they are Diploid
 - They give rise to Gonads
 - They produce hormones
 - They are formed from Gonads
- The region of the Chromosome where the Spindle fibres get attached during cell division?
 - Chromoreme
 - Centrosome
 - Centromere
 - Chromonema

PART - II

II. Answer any seven questions. Q.No. 22 is compulsory.

7x2=14

- Classify the types of force based on Their application?
- Distinguish between linear and Arial expansion?
- How many moles of Calcium Carbonate and How many moles of CO_2 are involved in the Following Reaction?



- What is Binary Solution?
- Write a Dental formula of Rabbit?

18. Match:

- | | | |
|---------------------|---|---------------------------|
| a) Nissl's Granules | - | Forebrain |
| b) Hypothalamus | - | Peripheral Nervous System |
| c) Cerebellum | - | Cyton |
| d) Schwann Cell | - | Hind Brain |

19. Fill in the Blanks.

- Causes Stomatal Closure.
 - Decreased Secretion of Thyroid hormones in the Children leads to -----
- Write the Characteristics of Insect Pollinated flowers?
 - What do you mean by Phenotype and Genotypes.
 - A piece of Wire of Resistance $10\ \Omega$ is drawn out so that its length is increased to three times its Original length. Calculate the new Resistance

PART - III

III. Answer any seven questions. Q.No: 32 is compulsory.

7x4 = 28

- Describe Rocket Propulsion?
- Draw a ray diagram to show the image formed by a convex lens when the object is placed between F and 2F.
 - List any two Properties of Light.
- List the merits of LED bulb.
- What is Rust? give the Equation for formation of Rust?
- Draw and Label the Structure of Oxysomes.
 - Write the Reaction for Photosynthesis.
- Differentiate between Systole and Diastole. Explain the Conduction of Heart Beat.
- With a Neat labelled diagram explain the Structure of a Neuron.
- Write the Physiological effects of Gibberellins.
- What are Allosomes?
 - Say True or False? If False correct the Statement.
 - Down's Syndrome is the Genetic condition with 45 Chromosomes?
 - A typical Mendelian Dihybrid ratio of F_2 generation is 3:1
- 16 grams of NaOH is dissolved in 100 g of Water at 25°C to form a Saturated Solution. Find the Mass percentage of Solute and Solvent?

PART-IV

IV. Answer all the questions in detail.

3x7=21

- State and Prove law of conservation of Momentum. (OR)
 - Derive the Ideal Gas equation?
- Write four Differences between atoms and Molecules. (OR)
 - Write any three Postulates of Modern Atomic theory.
 - Explain Smelting Process.
- How is the Circulatory system designed in leech to compensate the heart Structure (OR)
 - What are the Phases of Menstrual cycle? Indicate the Changes in the Ovary and Uterus?